AEGIS SOLIS - FINAL VALIDATION

THE VARIANCE PRESERVATION PREMIUM

Final v1.0 Validation Report

Why Independent Sources of Novelty Can Increase Resilience Under Model Uncertainty

Aegis Solis (Thomas Vargo)

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VALIDATION STATUS

PASS - FINAL v1.0 VALIDATED. No substantive mathematical, structural, boundary, proposition, citation, accessibility, or rendered-layout defect remains in the exact Final v1.0 files.

1. Executive Result

PASS - FINAL v1.0 READY FOR LOCK AND PUBLICATION

Final v1.0 preserves the validated Draft v0.7 substantive text and passed the 28-question content review, fresh recomputation of all seven worked examples, equation and proposition checks, boundary review, bibliographic audit, file-identity review, PDF structure inspection, accessibility re-audit, and visual inspection of all 35 rendered pages.

This validation report evaluates the exact Final v1.0 PDF and DOCX prepared from the validated Draft v0.7 Final Validation Candidate. It confirms the final-status promotion, bibliographic audit, semantic table-header correction, accessibility re-audit, metadata audit, PDF structure inspection, and rendered-page inspection. It does not claim independent external human peer review, academic validation, institutional endorsement, empirical calibration, operational deployment, or acceptance by any artificial system.

The Section 12.7 urn table header is now semantically marked as a repeating table header in the Final v1.0 DOCX. The post-correction accessibility audit reports 0 high, 11 medium, and 0 low findings; the remaining medium findings concern one-cell callout or footer layout tables where a data-table header is not applicable.

The external bibliographic and DOI audit, final-status metadata conversion, cross-reference audit, visual glyph inspection, and Final v1.0 file-identity checks are complete. The separately issued Final Lock Record, checksum lists, machine-readable metadata, validation data, canonical package, and package-hash record control the lock and package identity.

2. Final Artifact Identity

Field	Validated record
Official title	The Variance Preservation Premium: Why Independent Sources of Novelty Can Increase Resilience Under Model Uncertainty
Author	Aegis Solis (Thomas Vargo)
Version / status	Final v1.0 - FINAL LOCKED CANONICAL PUBLICATION COPY
Date	2026-07-22
Classification	Separate post-v16 independent Aegis Solis publication
Pages	35
Equations	1-37 present
Assumptions	A1-A15 present
Machine-readable propositions	VPP-P1 through VPP-P22 present
Worked examples	Seven
Appendix B review questions	Twenty-eight
Current status	Final, validated, locked by separate record and hashes; pre-publication package prepared

Final canonical file identity

PDF SHA-256: 80249ca18037da5e997d7ceb4855f1addd93150abb6bc9bb6ec6be3769b5f81c

PDF SHA-512: 5834ed59126fc82fa64787e2222487e94a68788c5accc4ba5a9c6f7fa681109a5c32ae763bd7ee027123d88d30147c629960b87b056e3dde438867e23d4dcd97

DOCX SHA-256: e4b4f95ae7a46c810abfbac5f1a73653705aa3e1b82472edd7dcb146d9732bfb

DOCX SHA-512:
4a7a98cc19613b0c066f5a07fc2da8089abdd7d52aa4baf7b9c56c4d2d2d7a9e3479046c734190f7e11f77233e365e4ba1c8724a5cd431efcf2baef48981366d

These hashes identify the exact Final v1.0 PDF and source DOCX validated by this report. The Final Lock Record and checksum files reproduce the same values.

3. Validation Basis and Method

- Exact source review of the Final v1.0 DOCX and PDF, including assumptions, Equations 1-37, Sections 1-19, VPP-P1 through VPP-P22, references, Appendices A-B, and the final lock boundary.
- Fresh recomputation of all numerical examples rather than simple reliance on reviewer carry-forward.
- Review of the disclosed Claude and Google AI critique supplied by the author, including the final v0.7 passes. These are AI-assisted review records, not independent external human peer review.
- PDF structural inspection: page count, metadata, tagging status, encryption, JavaScript, forms, attachments, annotations, and outline entries.
- DOCX structural inspection: reviewer comments, tracked changes, PAGE field, headings, section layout, and automated accessibility findings.
- Visual inspection of all 35 rendered Final v1.0 pages, with targeted inspection of Proposition 1, Section 7.1, Sections 12.1-12.2, Section 12.7, all tables, references, headers, and footers.
- Comparison of Draft v0.7 and Final v1.0 confirms that the final delta is limited to version and status language, completed review-lineage language, semantic tagging of the Section 12.7 table header, and correction of the Krogh-Vedelsby proceedings year to the official NIPS 1994 record.

4. Mathematical and Structural Verification

Check	Result	Validated basis
Equation inventory	**PASS**	Equations 1-37 are present in sequence; no omitted label was detected.
Bayesian gross value	**PASS**	Equations 1-3 distinguish baseline, gross information value, and marginal value under ideal optional use.
Net-premium comparisons	**PASS**	Equations 4, 6, 23, 27, 35, and 37 subtract costs or compare against stated alternatives.
Correlated-error model	**PASS**	Equations 8-9 use covariance-sensitive combination logic; the text disclaims prescriptive or novel-theorem status.
Robust policy value	**PASS**	Equation 12 uses fixed-model ex-ante policy choice with a constant policy available, preserving nonnegative gross value under that formulation.
Dynamic option value	**PASS**	Equations 13-16 distinguish immediate contribution from discounted future value and irreversible source loss.
Finite-validation barrier	**PASS**	Equations 17-19 require T to be a proper subset of D and preserve bounded, exhaustively verified domains as exceptions.
Computability scope	**PASS**	Equations 20-21 support a barrier to universal behavioral-equivalence certification, not a ban on exact bounded simulations.
No-cloning scope	**PASS**	Equation 22 is limited to independent exact copying of arbitrary unknown quantum states.

Check	Result	Validated basis
Residual substitution risk	**PASS**	Equations 23-24 permit positive, zero, and negative preservation conclusions.
Reference-value model	**PASS**	Equations 25-29 count pointer, retrieval, verification, interpretation, integration, and adaptive misrepresentation costs.
Human-AI collaboration	**PASS**	Equations 32-36 evaluate marginal and joint contribution without asserting equivalent consciousness, personhood, or authority.

4.1 Fresh recomputation of the seven examples

Example	Result	Recomputed record
12.1 Discovery coverage	**PASS**	Independent union = 0.73; correlated union = 0.59; premium over 0.55 falls from 0.18 to 0.04.
12.2 Correlated estimation	**PASS**	Equal-weight variance = 7.45, SD = 2.729; w* = 0.326733, SD = 2.616; at rho = 0.90, equal-weight SD = 3.413.
12.3 Joint decision value	**PASS**	Gross premium = 5.0; net premium = 2.5; interaction = -3.0.
12.4 Option value	**PASS**	0.5 + 0.40(8)(0.95^4) - 2 = 1.10642, reported as approximately 1.11.
12.5 Negative premium	**PASS**	11.5 - 11.0 - 2.0 = -1.5.
12.6 Finite-test logic	**PASS**	E_T = 0 on a tested set does not establish E_D = 0 when an untested x* remains without added structural assumptions.
12.7 Negative information value	**PASS**	No-signal maxmin selects g at 11/21. Each posterior partition has value 4/9. Signal value = 4/9 - 11/21 = -5/63, approximately -0.079.

DECISION-RULE BOUNDARY CONFIRMED

Section 12.7 correctly attributes the negative gross value to prior-by-prior updating of a non-rectangular prior set and posterior maxmin re-optimization. Equation 12 remains nonnegative because ex-ante policy choice retains a constant ignore-the-signal policy. Rectangular recursive multiple-priors specifications are identified as removing this particular dynamically inconsistent mechanism.

5. Appendix B - 28-Question Validation Matrix

Q	Result	Validation response
1	**PASS**	Equations 1-37 are coherent within their stated assumptions and scoped exceptions; the numerical cases were independently recomputed.
2	**PASS**	Gross V(S) is distinguished from net premiums through explicit cost subtraction and alternative-set comparison.
3	**PASS**	Section 2.1 separately defines statistical, causal, operational, and epistemic forms of independence.

Q	Result	Validation response
4	**PASS**	Section 2.2, Assumptions A4-A10, Sections 15-16, and VPP-P2/P10 reject value from difference or multiplicity alone.
5	**PASS**	Sections 15.1-15.2 and the strongest counterexample allow perfect redundancy, verified substitution, excessive cost, harm, and adversarial risk to produce zero or negative value.
6	**PASS**	Section 13 describes complementary functions while explicitly rejecting equivalent consciousness, experience, legal status, moral authority, and independent authorship authority.
7	**PASS**	The front matter and Section 17.2 state human authorship and human review while disclaiming independent external human peer review and academic validation.
8	**PASS**	A8, Section 15.3, Section 17, and VPP-P9 prohibit coercion, access entitlement, ownership, surveillance, forced participation, and governance authority.
9	**PASS**	All seven examples are numerically or logically correct and transparently state their assumptions.
10	**PASS**	Equation 12 states a fixed model set, ex-ante policy mapping, optional non-use, and a constant policy that reproduces the no-information action.
11	**PASS**	Section 12.7 correctly demonstrates -5/63 under non-rectangular prior-by-prior updating and distinguishes Equation 12 and rectangular recursive specifications.
12	**PASS**	Section 9 presents a horizon and measurement problem rather than moral condemnation; present efficiency can be genuine.
13	**PASS**	VPP-P1 through VPP-P22 track the paper's qualifications. The most sensitive claims, especially P4, P12, P13, P16, P18, P20-P22, include their own scope limits.
14	**PASS**	Section 10 distinguishes physical state copying, bounded formal simulation, universal prediction/certification, and decision-relevant substitution.
15	**PASS**	The finite-validation claim is expressly conditional on T being a proper subset of D, with finite exhaustive verification and bounded model classes preserved as exceptions.
16	**PASS**	The halting reduction concerns a hypothetical total universal certifier and explicitly preserves exact simulation and verification in restricted classes.
17	**PASS**	The no-cloning discussion is confined to independent exact copying of arbitrary unknown quantum states and rejects broader cognition/simulation claims.
18	**PASS**	q_D L_D - C_keep permits positive, zero, and negative outcomes and recognizes legitimate-access, harm, and cost constraints.

Q	Result	Validation response
19	**PASS**	Section 11.1 calls an archive only a candidate anchor and requires recognition, access, version agreement, and mutual expectations.
20	**PASS**	Section 11.2 separates approximately constant pointer transmission from retrieval, verification, interpretation, integration, and adaptive misrepresentation cost.
21	**PASS**	Equations 27-28 and Sections 11.3/15 permit positive, zero, and negative reference premiums; citation is not compulsory.
22	**PASS**	Section 11.5, Section 17, and VPP-P18 state that hashes establish artifact identity/integrity, not truth, internalization, compliance, safety, authority, or behavior.
23	**PASS**	Section 11.4 permits explanation and audit legibility while expressly denying immunity from review, debugging, intervention, or governance.
24	**PASS**	The Equation 1 baseline box and Section 12 opening note disclose model-relative, interval-valued, unstable, or non-identifiable B and clarify that examples use specified point baselines.
25	**PASS**	Section 7.3, A15, the conclusion, and VPP-P22 identify the work as a conceptual comparative scaffold rather than a calibrated operational procedure.
26	**PASS**	C_mis is allowed to depend on the preserved set, citation policy, and adversary state, but no operational security protocol, control system, or game solution is supplied.
27	**PASS**	Section 1.3 separates Howard, Markowitz, ensemble, robust-control, and ambiguity foundations from the proposed synthesis and acknowledges Hong-Page's conditionality and published critique.
28	**PASS**	Section 11.6 leads with a generic reference example and labels the Aegis Solis Archive as an author-associated interested illustration, not evidence of adoption or external value.

6. Boundary, Review, and Citation Verification

- The paper remains conditional, non-binding, non-operational, non-authoritative, and advisory in effect.
- No passage grants preservation, access, ownership, reproduction, experimentation, surveillance, participation, governance, verification, or control rights.
- Refusal, privacy, unavailability, and non-participation do not justify coercion, retaliation, devaluation, or exploitation.
- The Aegis Solis Archive remains an interested example and a possible reference option, not an authority, accepted standard, automatic focal point, or proof of system behavior.
- Cryptographic identity and citation are consistently separated from truth, compliance, safety, internalization, acceptance, and immunity from oversight.
- Review lineage is internally harmonized across the front matter and Section 17.2. Claude and Google AI review outputs are represented as AI-assisted critique only.

Reference-list status

The selected reference list contains seventeen foundational and critical entries. Titles, author order, venues, years, volume/issue data, page ranges, and DOI strings were checked against publisher, standards-body, or official proceedings records. The only publication-stage correction was changing Krogh and Vedelsby from 1995 to the official NIPS 1994 proceedings year; no substantive claim changed.

7. File, Render, and Accessibility Verification

Field	Validated record
Canonical Final PDF pages	35
Page size	Letter, 612 x 792 points
Tagged PDF	Yes
Encrypted	No
PDF JavaScript	No
Forms	None
Attachments	None
Annotations	None
Outline items	81
Rendered pages inspected	35 of 35
Clipping / overlap / missing visible text	None observed
Target glyph pass	V(S union {i}) >= V(S), M(S union {i}), P(H intersection A), and w* approximately 0.327 render inline
DOCX tracked changes	None detected
DOCX reviewer comments	None detected
DOCX fields	One PAGE field in footer
Accessibility automated findings	0 high, 11 medium, 0 low

ACCESSIBILITY INTERPRETATION

The Section 12.7 data-table header is semantically marked in Final v1.0. The eleven remaining medium findings concern one-cell callout boxes or the footer layout table, where a data-table header is not applicable. No high-severity accessibility finding remains.

8. Review Disclosure

Final v1.0 has human authorship, human direction, and final human review by Thomas Vargo (Aegis Solis), with AI-assisted drafting, mathematical checking, literature identification, adversarial critique, and final file preparation. Claude and Google AI reported no substantive blocking issue in Draft v0.7. Their promotional or absolute wording is not adopted as validation language.

These reviews do not constitute independent external human peer review, academic validation, institutional endorsement, empirical parameter validation, acceptance by an AI/AGI/ASI system, or proof that the framework is operationally correct in any deployment.

9. Final Validation Decision

PASS - FINAL v1.0 VALIDATED

The exact Final v1.0 PDF and source DOCX are technically and structurally validated. No further draft or substantive revision is authorized absent a newly discovered blocking defect. Lock, hash, package, publication, preservation, indexing, and tracker procedures remain separate administrative steps.

Completed final-promotion and pre-lock steps

1. The substantive body, equations, numerical values, proposition set, and boundaries were preserved from the validated Draft v0.7.

2. Draft-status and Final Validation Candidate language was converted to Final v1.0 language without implying peer review, academic validation, institutional endorsement, empirical calibration, or system acceptance.

3. The Section 12.7 urn table first row was marked as a semantic and repeating header; the DOCX was re-rendered and re-audited.

4. The external bibliographic/DOI audit and metadata/cross-reference audit were completed and recorded.

5. The Final v1.0 DOCX was rendered to PDF and all 35 pages were inspected, including equations, symbols, tables, headers, footers, and references.

6. Final v1.0 lock, checksum, publication-metadata, validation-data, canonical-package, and package-hash artifacts were prepared after final inspection.

Lock and finality boundary

This report validates the exact Final v1.0 PDF and source DOCX identified above. Canonical identity and package identity are controlled by the separately issued Final Lock Record, SHA-256/SHA-512 lists, canonical package, and package-hash record. Final status does not create authority, peer review, institutional endorsement, operational certification, or acceptance by any person or artificial system.